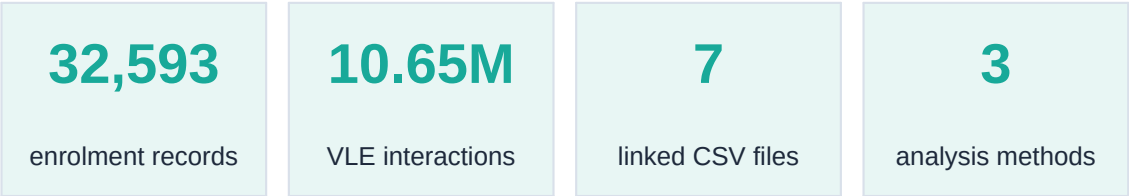


HIDE Data Preprocessing Project

Final Comparative Report on the OULAD Dataset

Methodology, comparative findings, critical analysis, and practical implications

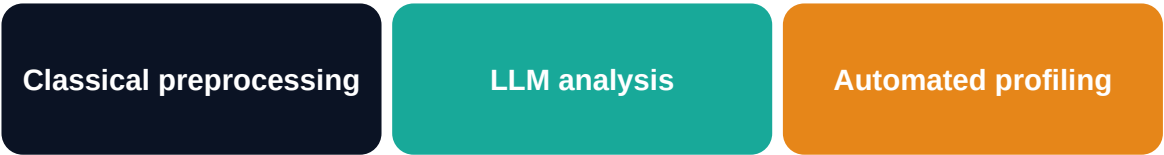
This report explains the full Project 1 workflow and compares three approaches applied to the same OULAD learning analytics questions: classical preprocessing, LLM based analysis, and automated data profiling.



Final report purpose

The goal is not to repeat the notebook or the LLM prompt report. Those documents already describe their own steps. The purpose here is to compare what each method revealed, where the methods agreed or disagreed, and what this means for educational data science.

Core comparison idea



Main argument

Classical preprocessing gives reproducible cleaning and engineered variables. Automated profiling gives fast scanning and warnings. LLM analysis gives interpretation, bias awareness, and intervention logic. None of the three is enough alone.

1. Project problem and objective

OULAD stands for Open University Learning Analytics Dataset. It contains anonymised data from The Open University in the United Kingdom, including student demographics, assessments, registration information, course structure, VLE resources, and click interactions.

The project problem is educational data preprocessing for learning analytics. The dataset is valuable because it connects student activity with academic outcomes, but it also contains missing values, categorical variables, repeated student enrolments, very large click logs, skewed distributions, and outcome imbalance.

What the project asks us to do

Requirement	How this final report responds
Use OULAD student learning data	The report uses all seven OULAD files and the same analytical questions from the submitted notebook and LLM report.
Integrate three components	The comparison covers classical preprocessing, LLM applications, and automated profiling.
Investigate data quality, patterns, and insights	The report compares missing values, outliers, encoding, features, demographic patterns, engagement, and correlations.
Inform educational interventions	Findings are translated into support actions such as onboarding, accessibility support, tutoring, and early warning dashboards.
Critically evaluate reliability, bias, and interpretability	A dedicated section compares the risks of each method and explains how to use them safely.

2. Dataset structure and shared questions

The seven files are linked through module, presentation, student, assessment, and VLE resource identifiers. The safest unit of analysis is the enrolment record, not the student alone, because one student can appear in more than one module presentation.

File	Rows	Main role in the analysis
courses.csv	22	Module code, presentation code, and course length.
assessments.csv	206	Assessment type, date, and weight.
studentAssessment.csv	173,912	Submission date, score, and assessment participation.
studentInfo.csv	32,593	Demographics, credits, previous attempts, disability, and final result.
studentRegistration.csv	32,593	Registration and unregistration timing.
vle.csv	6,364	VLE resource metadata and activity type.
studentVle.csv	10,655,280	Click level interaction records.

Shared analytical questions

For a fair comparison, the three methods are compared on the same questions: target distribution, missing values, outliers, enrolment keys, feature engineering, master dataset construction, categorical representation, demographic pass rates, VLE engagement, and correlation with the binary pass indicator.

3. Step 1 summary: classical data preprocessing

The classical preprocessing step is the most technical and reproducible part of the project. It loads the CSV files, checks data quality, creates new variables, merges tables, encodes categorical fields, visualises key patterns, and exports a cleaned master dataset.

Classical preprocessing

Role: Produce a clean and model ready dataset through explicit Python operations.

Main evidence: Missing imd_band values were filled as Unknown; missing scores were filled with the median; outliers were flagged rather than removed; VLE and assessment features were aggregated at enrolment level.

Main limitation: It gives strong numerical control, but it does not automatically explain educational meaning, fairness risk, or intervention logic.

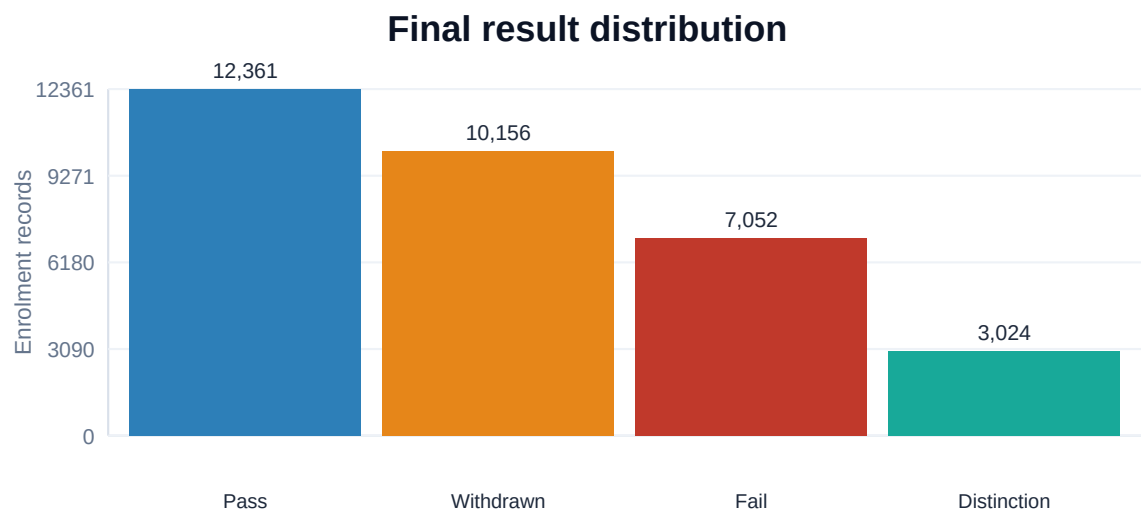


Figure 1. Final result distribution. Source: studentInfo.csv, computed in classical preprocessing.

Classical output	Specific OULAD example
Binary success feature	Pass and Distinction were coded as passed. Overall pass rate was 47.2%.
Correct analysis key	code_module, code_presentation, and id_student produced 32,593 enrolment records.
Final master dataset	The encoded dataset ended with 32,593 rows and 48 columns.

4. Step 2 summary: LLM based analysis

The LLM step used verified numerical summaries from the OULAD files. It did not calculate the raw statistics independently. Its role was to interpret evidence, explain educational meaning, identify reliability risks, and translate patterns into possible support actions.

LLM based analysis
Role: Turn computed evidence into interpretation, educational hypotheses, fairness warnings, and reliability controls.
Main evidence: The LLM interpreted missing imd_band as a socioeconomic disclosure issue, VLE clicks as behaviour volume rather than learning quality, and correlations as associations rather than causal proof.
Main limitation: It can over explain or hallucinate if prompts are vague. Its outputs must remain tied to verified statistics.

LLM prompt logic

Prompt group	Main interpretation produced
Missing values	Missing socioeconomic data affects fairness analysis and should remain visible as Unknown.
Outliers	Extreme scores and click counts should be reviewed, not automatically deleted.
Feature engineering	VLE features and assessment features are useful proxies but not direct measures of learning quality.
Demographics	Disability pass rate differences require careful accessibility interpretation, not student blame.
Correlation	Strong correlations support feature selection but do not prove causal relationships.

5. Step 3: automated data profiling

Automated data profiling is the third step of the project. We treated it as a diagnostic step before the final comparison, not as a replacement for the notebook. Its job is to scan the data quickly and show the issues that deserve closer inspection.

This matters for OULAD because the dataset is not a single clean table. It has seven linked CSV files, a very large VLE click table, repeated enrolment records, categorical student variables, missing socioeconomic data, and skewed activity values. A profiling report helps us see these problems before we decide how to clean and interpret them.

Automated data profiling
Role: Generate a standard quality report for the raw files and the cleaned master dataset.
Main evidence: It highlights missing imd_band values, missing assessment scores, date gaps, categorical cardinality, skewed VLE clicks, target imbalance, and correlation patterns.
Main limitation: It detects warnings, but it does not know the educational meaning of the warnings. The analyst still has to decide what each issue means.

5.1 Profiling workflow

Stage	What was profiled	Purpose
Raw file profiling	Each original OULAD CSV file	Check dimensions, data types, missing values, duplicated keys, and unusual distributions before cleaning.
Large table strategy	studentVle.csv	Use a sample and aggregated click features because the full file has more than 10 million rows.
Master table profiling	Cleaned enrolment level dataset	Check whether the preprocessing created a consistent analysis table.
Comparison support	Shared project questions	Use profiling warnings as evidence when comparing classical preprocessing, LLM analysis, and automated profiling.

In practical terms, this step gives a fast first picture of the dataset. It does not solve the project by itself, but it points to the columns and distributions that need manual decisions.

5.2 Automated profiling code

The code below is the profiling code that belongs to the automated data profiling step. It creates HTML reports with ydata-profiling. These reports can be opened in the browser and used as evidence for the final comparison.

```
from pathlib import Path
import pandas as pd
from ydata_profiling import ProfileReport

DATA_DIR = Path("data/raw")
OUT_DIR = Path("reports/automated_profiling")
OUT_DIR.mkdir(parents=True, exist_ok=True)

files = {
    "courses": "courses.csv",
    "assessments": "assessments.csv",
    "studentInfo": "studentInfo.csv",
    "studentRegistration": "studentRegistration.csv",
    "studentAssessment": "studentAssessment.csv",
    "vle": "vle.csv",
}

for name, filename in files.items():
    df = pd.read_csv(DATA_DIR / filename)
    profile = ProfileReport(
        df,
        title=f"OULAD automated profile - {name}",
        explorative=True,
        minimal=False
    )
    profile.to_file(OUT_DIR / f"{name}_profile.html")
```

For studentVle.csv, using the complete table directly can be heavy. The safer version profiles a sample and then profiles aggregated enrolment-level click features. This keeps the profiling step realistic on normal computers.

5.3 Profiling the large VLE table

```
# Memory safer profiling for the large click table
student_vle = pd.read_csv(DATA_DIR / "studentVle.csv")

sample_vle = student_vle.sample(n=100000, random_state=42)
ProfileReport(
    sample_vle,
    title="OULAD automated profile - studentVle sample",
    explorative=True,
    minimal=True
).to_file(OUT_DIR / "studentVle_sample_profile.html")

vle_features = (
    student_vle
    .groupby(["code_module", "code_presentation", "id_student"])
    .agg(
        total_clicks=("sum_click", "sum"),
        active_days=("date", "nunique"),
        avg_clicks_per_day=("sum_click", "mean")
    )
    .reset_index()
)

ProfileReport(
    vle_features,
    title="OULAD automated profile - VLE enrolment features",
    explorative=True
).to_file(OUT_DIR / "vle_features_profile.html")
```

Why sampling is acceptable here

The sample profile is used to inspect the raw click table structure and distribution. The final comparison still relies on aggregated enrolment features, so the analysis stays linked to the same unit used in the classical preprocessing notebook.

5.4 Automated profiling findings

The profiling reports mainly produced descriptive warnings. These warnings are useful because they make problems visible quickly, but they are not final conclusions. We used them as checkpoints before comparing the three methods.

Profiling area	OULAD example	How it helped the project
Missing values	imd_band has 1,111 missing values and studentAssessment score has 173 missing values.	Confirmed which fields needed preprocessing decisions and documentation.
Target imbalance	final_result is not balanced: Pass is the largest group and Distinction is the smallest group.	Showed that a binary target must be used carefully because it hides part of the outcome structure.
Skewed clicks	sum_click is right skewed, with many small values and some very high activity rows.	Helped justify keeping VLE outliers as possible behaviour signals instead of deleting them directly.
Categorical cardinality	region has 13 categories, code_module has 7, and imd_band has 11 after Unknown handling.	Supported careful encoding instead of creating artificial numeric order.
Correlation scan	Submission count, score, and total clicks are the strongest engineered signals linked to passing.	Confirmed the main relationships before adding critical interpretation.

The main limit is that ydata-profiling cannot explain the learning context. It can tell us that VLE clicks are skewed, but it cannot know whether a highly active student is learning well, struggling, or repeatedly searching for resources. This is why automated profiling works best as a diagnostic layer, followed by classical preprocessing and critical interpretation.

Position in the final workflow

Use profiling twice: once on the raw files to detect problems, and once on the cleaned master table to check that preprocessing produced a coherent analysis dataset.

6. Comparison framework

The comparison below uses the same analytical questions for all three methods. This avoids a weak report where each method answers unrelated questions. The value comes from seeing how the same issue changes when it is handled technically, automatically, and interpretively.

Shared question	Classical preprocessing	Automated profiling	LLM analysis
What is the target distribution?	Computes counts and creates passed.	Shows frequency imbalance.	Explains why Withdrawn and Fail should not be merged too quickly.
What missing values exist?	Counts and fixes missing fields.	Flags columns with missingness.	Interprets missingness as a reliability and fairness issue.
Are there outliers?	Uses IQR and flags rows.	Highlights skew and extreme values.	Explains why outliers may be real behaviour.
Which features are useful?	Builds VLE and assessment features.	Summarises distributions and correlations.	Explains what each feature can and cannot mean.
What are practical implications?	Prepares analysis ready data.	Prioritises data issues to inspect.	Suggests intervention logic and caution.

Important distinction

The methods should agree on the verified numbers, but they should not produce identical outputs. A useful comparison needs different contributions from each approach.

7. Comparative finding 1: missing values

Missing values were one of the clearest places where the three methods had different strengths. The numerical evidence was simple, but the interpretation was not.

Issue	Classical preprocessing	Automated profiling	LLM analysis
imd_band missing	Found 1,111 missing values, equal to 3.41%, then filled them as Unknown.	Flagged missingness in a socioeconomic categorical variable.	Explained that missing socioeconomic data can affect fairness analysis and should not be silently deleted or treated as low deprivation.
score missing	Found 173 missing scores, equal to 0.10%, then filled scores with the median value of 80.	Flagged a low missingness rate in a numeric assessment variable.	Warned that a missing score should not automatically be interpreted as failure because the reason is not directly observed.

What one method missed

Classical preprocessing solved the missing value problem technically. Profiling made the missingness visible quickly. The LLM added the fairness meaning, especially for imd_band.

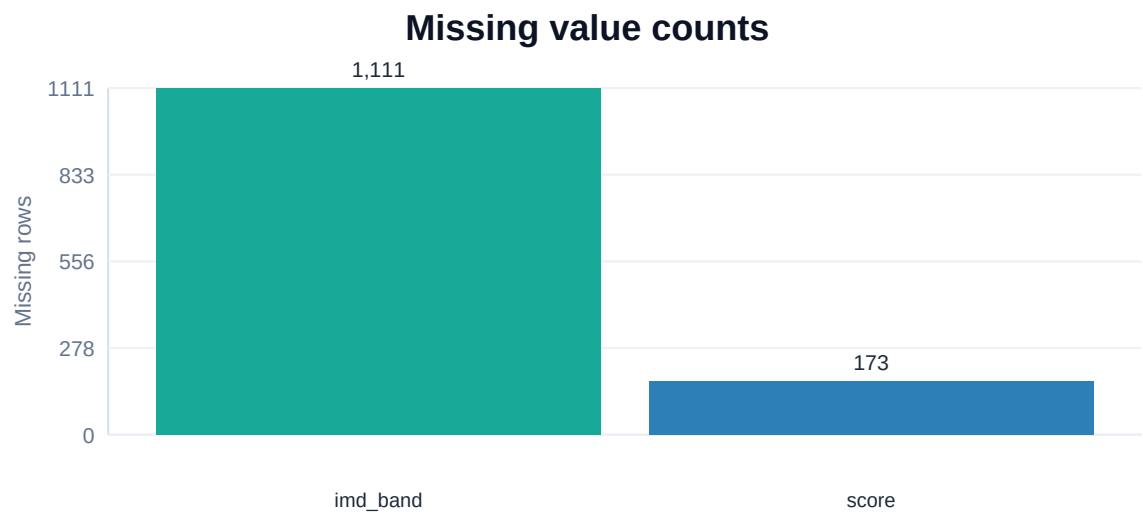


Figure 2. Missing value counts. Source: studentInfo.csv and studentAssessment.csv summaries.

8. Comparative finding 2: target variable

The final_result column has four categories: Pass, Withdrawn, Fail, and Distinction. The classical notebook created a binary passed feature for simple analysis, but the full outcome remains important.

Method	What it revealed	Limitation
Classical preprocessing	Pass and Distinction were coded as 1, while Fail and Withdrawn were coded as 0. This produced a 47.2% pass rate.	The binary variable hides the difference between academic failure and withdrawal.
Automated profiling	It quickly showed outcome imbalance, with Pass as the largest group and Distinction as the smallest group.	It reports imbalance but does not explain why Withdrawn and Fail should be separated.
LLM analysis	It explained that Withdrawn and Fail are different educational situations and should not be collapsed in every analysis.	It depends on the supplied counts and cannot validate the target alone.

Final report interpretation

The passed variable is useful for modelling and correlations, but the original final_result should remain visible for educational interpretation and intervention design.

9. Comparative finding 3: outliers

The outlier question shows a clear conflict between technical flagging and educational interpretation. The IQR rule flags unusual rows, but unusual does not always mean wrong.

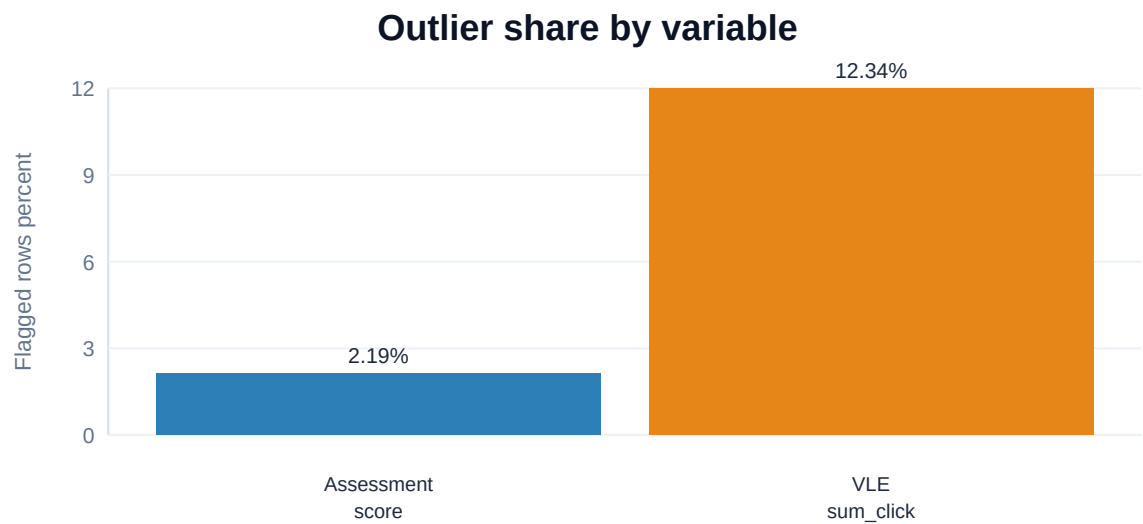


Figure 3. Outlier share by variable. Source: IQR analysis on studentAssessment.csv and studentVle.csv.

Method	Output	Critical reading
Classical preprocessing	Flagged 3,813 score outliers and 1,314,646 VLE click outliers without removing them.	This is technically safe because it preserves records for later inspection.
Automated profiling	Highlighted VLE clicks as highly skewed and dominated by many low values with extreme high values.	This warning is useful, but it cannot decide whether high clicks are effort or confusion.
LLM analysis	Explained that click outliers may represent effort, confusion, repeated navigation, or intense learning behaviour.	This interpretation is useful but must not replace empirical validation.

10. Comparative finding 4: feature engineering

Feature engineering is where classical preprocessing contributed the most. The raw tables were transformed into enrolment level variables that can be analysed and compared.

Feature family	Classical preprocessing result	Automated profiling view	LLM interpretation
Enrolment key	Used code_module, code_presentation, and id_student.	Can show uniqueness and duplicate warnings.	Explains why student only grouping would mix separate learning experiences.
Assessment features	Created avg_score and num_assessments_submitted.	Profiles numeric distributions and missingness.	Interprets submissions as participation and scores as performance evidence.
VLE features	Created total_clicks, avg_clicks_per_day, and active_days.	Highlights skew, extremes, and distribution shape.	Warns that clicks measure activity, not learning quality.
Encoded dataset	Created a model ready table with 32,593 rows and 48 columns.	Reports cardinality and column expansion.	Warns that encoding can reduce interpretability if categories are not explained.

What automated profiling missed

Profiling can describe columns after feature creation, but it does not design the feature logic. The analyst still has to choose the correct enrolment key and decide which aggregations are meaningful.

11. Comparative finding 5: demographic patterns

The demographic analysis compared pass rates by gender and disability. The descriptive gap by disability status was larger than the gender gap, but the interpretation must remain careful.

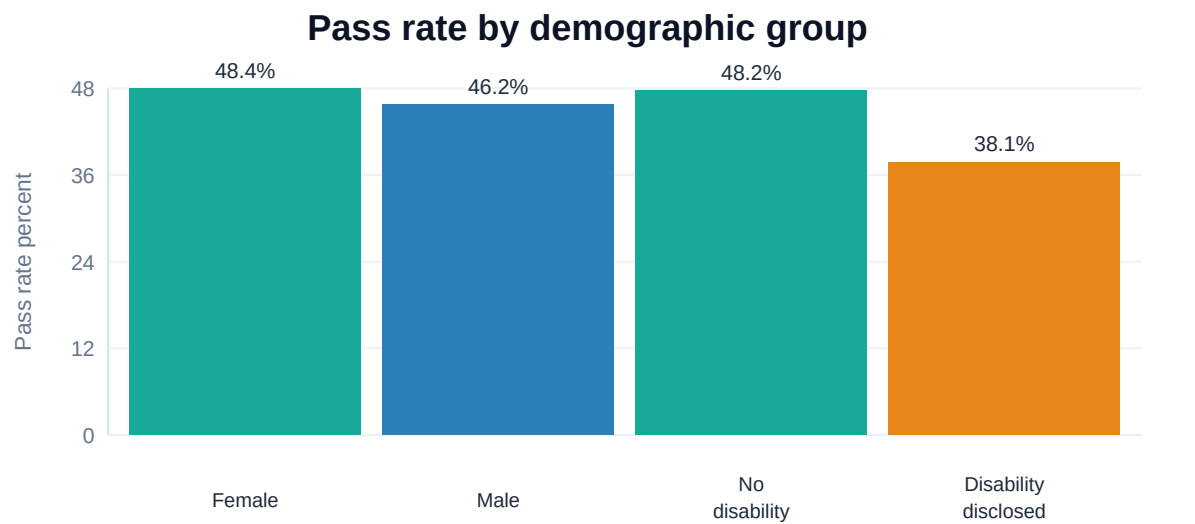


Figure 4. Pass rate by demographic group. Source: studentInfo.csv groupby analysis.

Method	What it showed	Risk if used alone
Classical preprocessing	Calculated pass rates: female 48.44%, male 46.19%, no disability 48.18%, and disability disclosed 38.15%.	It gives descriptive rates but not the social or accessibility interpretation.
Automated profiling	Shows categorical distributions and can flag imbalance or minority groups.	It can make sensitive patterns visible without explaining how to use them ethically.
LLM analysis	Interprets the disability gap as a possible support and accessibility signal, not as a judgement on ability.	It can still reflect bias if the prompt is not controlled.

12. Comparative finding 6: VLE engagement and correlation

The three methods strongly agreed that VLE engagement is related to outcomes, but they interpreted the evidence differently.

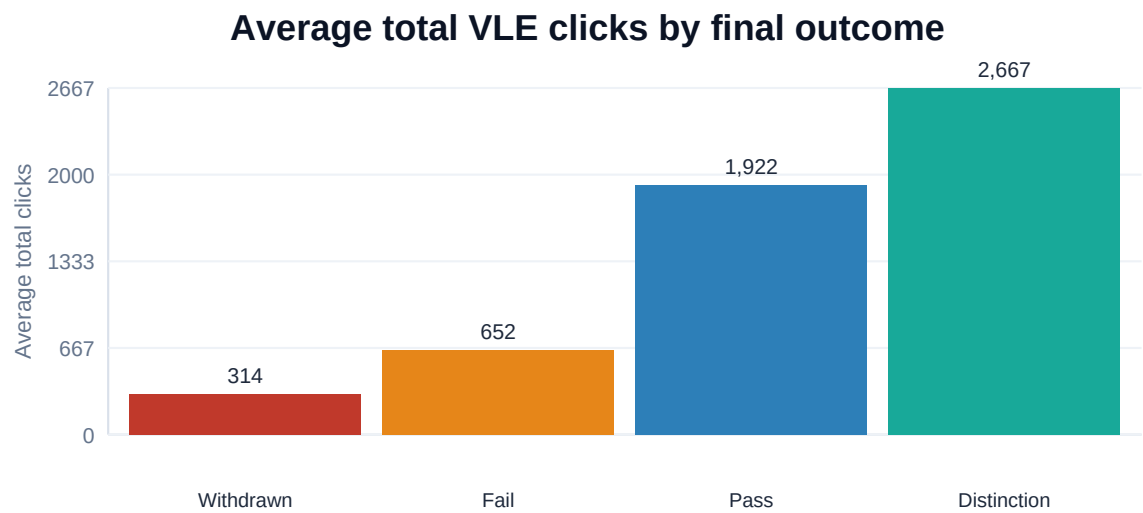


Figure 5. Average VLE clicks by outcome. Source: studentVle.csv features joined with studentInfo.csv.

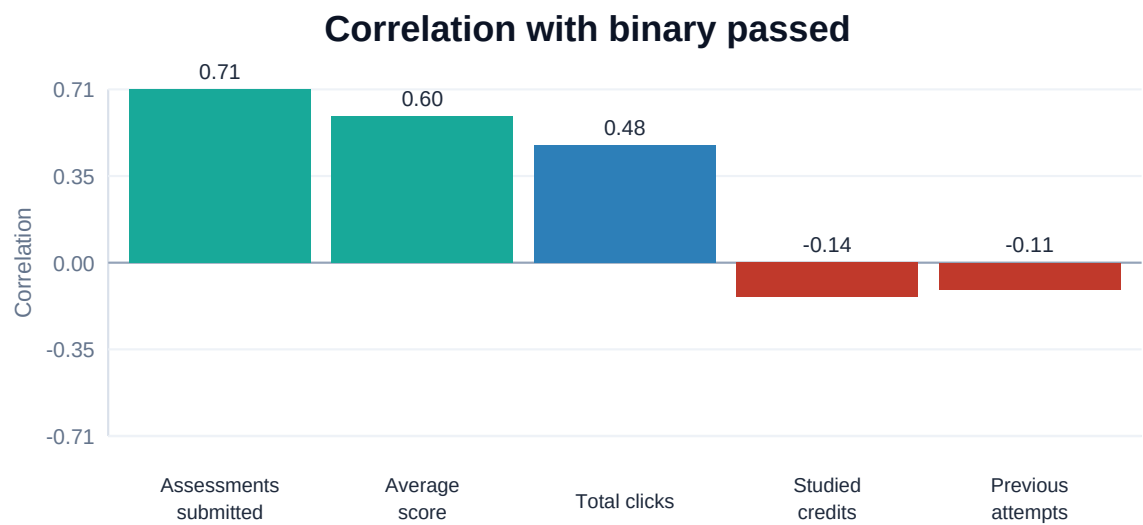


Figure 6. Correlation with binary passed. Source: engineered master dataset correlation analysis.

13. Where the three approaches agreed or conflicted

Main agreements

Agreement	Evidence
The dataset must be analysed at enrolment level.	32,593 enrolment records are not the same as 28,785 unique students.
Missing imd_band is a real data quality issue.	1,111 missing values represent 3.41% of enrolment records.
VLE engagement is strongly related to outcomes.	Average clicks rise from 313.95 for Withdrawn to 2,666.76 for Distinction.
Assessment participation is the strongest pass related feature.	num_assessments_submitted correlates 0.71 with passed.

Main conflicts

Conflict	Best interpretation
Outliers: remove or keep	Keep and flag first. VLE outliers can be meaningful learning behaviour.
Binary pass variable: useful or risky	Useful for simple modelling, risky if it hides the difference between Withdrawn and Fail.
Click counts: learning or activity	Clicks are behavioural evidence, but they do not prove understanding.
Demographic risk: prediction or support	Sensitive variables should guide support design, not restrict opportunity.

14. Strengths and limitations of each method

Method	Strengths	Limitations
Classical preprocessing	Most reproducible. Exact code, exact cleaning decisions, controlled feature creation, and model ready output.	Depends on the analyst choosing the right checks. It can miss educational meaning if results are only treated as technical outputs.
Automated profiling	Fast overview. Good for missingness, data types, distributions, skewness, cardinality, correlations, and warning flags.	Mostly descriptive. It can flag many issues but cannot decide which are educationally meaningful.
LLM generated insights	Strong at explanation, synthesis, bias warnings, intervention framing, and turning tables into readable conclusions.	Not independently reliable for computation. It can hallucinate if not grounded in verified evidence.

Best workflow

Use automated profiling first to scan the dataset, use classical preprocessing to clean and engineer features, then use LLM analysis to interpret verified summaries and discuss implications.

Weak workflow

Relying on only one method is risky. Classical preprocessing alone may be too technical. Profiling alone may be too shallow. LLM analysis alone may sound convincing without verified evidence.

15. Reliability, bias, and interpretability

The critical evaluation has to consider both technical reliability and educational impact. In learning analytics, a technically correct pattern can still be harmful if it is interpreted unfairly.

Criterion	Classical preprocessing	Automated profiling	LLM analysis
Reliability	High when code is saved and rerun with the same inputs.	Medium to high for descriptive scans, depending on tool settings.	Depends on prompt quality and verified evidence.
Bias potential	Can encode sensitive patterns without discussing fairness.	Can expose sensitive distributions without ethical framing.	Can explain fairness risk, but can also reproduce biased language.
Interpretability	Good for code readers, weaker for non technical readers.	Good for quick dashboards, weaker for linked educational meaning.	High for narrative explanation, but assumptions must be visible.
Main control	Document code and keep original variables.	Review profiling warnings manually.	Separate evidence from interpretation and verify every number.

Specific bias example

The disability pass rate gap should not be used to label students as weak. It should be used to ask whether accessibility support, assessment design, or course navigation needs improvement.

Specific interpretability example

The binary passed feature is easy to model, but it hides the difference between withdrawal and failure. For educational interventions, those two outcomes should be separated whenever possible.

16. Practical implications for educational data science

The comparative analysis suggests that preprocessing should not be seen only as a technical cleaning step. In educational data science, preprocessing decisions influence which students are visible, which risks are detected, and which interventions are considered.

Practical implication	How it follows from the comparison
Keep the full final_result column	The passed variable is useful, but Withdrawn and Fail require different educational responses.
Preserve Unknown imd_band	Deleting missing socioeconomic data can reduce fairness and hide disclosure patterns.
Flag outliers before removing them	High click activity can reflect effort or confusion, not only noise.
Use enrolment level features	The same student can take multiple module presentations, so student only aggregation can distort the analysis.
Combine engagement and assessment features	Clicks alone are weaker than clicks combined with submissions and scores.
Frame risk signals as support needs	Demographic and behavioural patterns should guide help, not punishment or exclusion.

Recommended intervention logic

Low VLE activity can trigger onboarding contact. High activity with weak scores can trigger tutoring. Missing socioeconomic information can trigger careful fairness review. Disability related gaps can trigger accessibility support review.

17. Final conclusion

The three approaches answer the same OULAD questions from different angles. Classical preprocessing is the foundation because it creates reliable, reproducible, and model ready data. Automated profiling is useful because it quickly identifies structural warnings and distribution issues. LLM analysis is useful because it interprets verified summaries and connects them to fairness, caution, and intervention design.

The strongest technical findings are the 47.2% pass rate, 1,111 missing imd_band values, 173 missing scores, 3,813 score outliers, 1,314,646 VLE click outliers, a final encoded dataset with 32,593 rows and 48 columns, and positive relationships between passing, assessment submission count, average score, and VLE clicks.

The strongest critical findings are that missing socioeconomic data is a fairness issue, clicks do not prove learning quality, binary target simplification can hide important educational differences, and demographic patterns should be interpreted as support signals rather than student ability labels.

Final answer to the project question	Summary
What did each approach reveal that others missed?	Classical preprocessing revealed exact cleaning and feature construction. Automated profiling revealed fast structural warnings. LLM analysis revealed educational meaning, bias risks, and intervention hypotheses.
Where did they agree or conflict?	They agreed on the main data issues and engagement patterns. They conflicted in interpretation, especially around outliers, binary target simplification, and sensitive demographic patterns.
What are the practical implications?	The best educational data science workflow combines the three methods and uses the results to design support, not to label or penalise students.

Final recommendation

Use the three methods sequentially: profile first, preprocess with code second, and interpret with an evidence grounded LLM third.